Janagani Shrethan Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	5.80
SR EDUCATIONAL ACADAMY	Class XII (TSBIE)	2024	949.00%
SPR SCHOOL	Class X (TS SSC)	2022	10.00%

SCHOLASTIC ACHIEVEMENTS

Received the Innovation Recognition Award at the MoPR AI/ML Hackathon, organized by the Ministry of Panchayati Raj and conducted by IIT Tirupati.

POSITIONS OF RESPONSIBILITY

Team Lead, Geo-Intel Hackathon Team (4 Members)

Led task coordination across LiDAR processing, hydrological modelling, cloud deployment, and visualization.

SKILLS

Programming Languages : Python, Verilog/SystemVerilog, MATLAB, C, Bash

Python Libraries : NumPy, Pandas, Scikit-learn, WhiteboxTools, GDAL,Rasterio, Folium

Machine Learning : Random Forest, Neural Networks, Pattern Recognition

Soft Skills : Technical Writing, Cross-functional Team Collaboration

RELEVANT COURSES

Pattern Recognition & Machine Learning, Neural Networks,

Machine Learning, Data Structures & Applications,

Digital Signal Processing, Signal Processing,

Control Systems, Power System Analysis,

Analog Electronics, Random Processes

PROJECTS

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Geo-Intel DTM Pipeline | 2026

- Developed an AI-powered pipeline to generate engineering-grade drainage plans from drone LiDAR point clouds.

- Built a Random Forest classifier and implemented hydrological modelling with FastAPI-based visualization.

Q-Learning for Phone Battery Management | 2026

- Modelled battery management as a Markov Decision Process and implemented tabular Q-Learning in Python.

- Optimized policy performance using -greedy exploration and hyperparameter tuning.

3-Layer Feedforward Neural Network for MNIST | 2026

- Developed a 3-layer neural network from scratch using NumPy for handwritten digit classification.
- Implemented backpropagation, Xavier initialization, and mini-batch SGD to improve model performance.

FPGA Design Projects

- Designed and implemented an 8-point Radix-2 FFT using Verilog on the DE10-SoC FPGA.
- Performed HPS bring-up and designed RC Phase Shift, Wien Bridge, and Schmitt Trigger oscillator circuits.

EXTRACURRICULAR

- Participated in the National Geospatial Intelligence Hackathon organized by MoPR and IIT Tirupati.